

Aprendizado Profundo

Modelos Multimodais - LLAVA

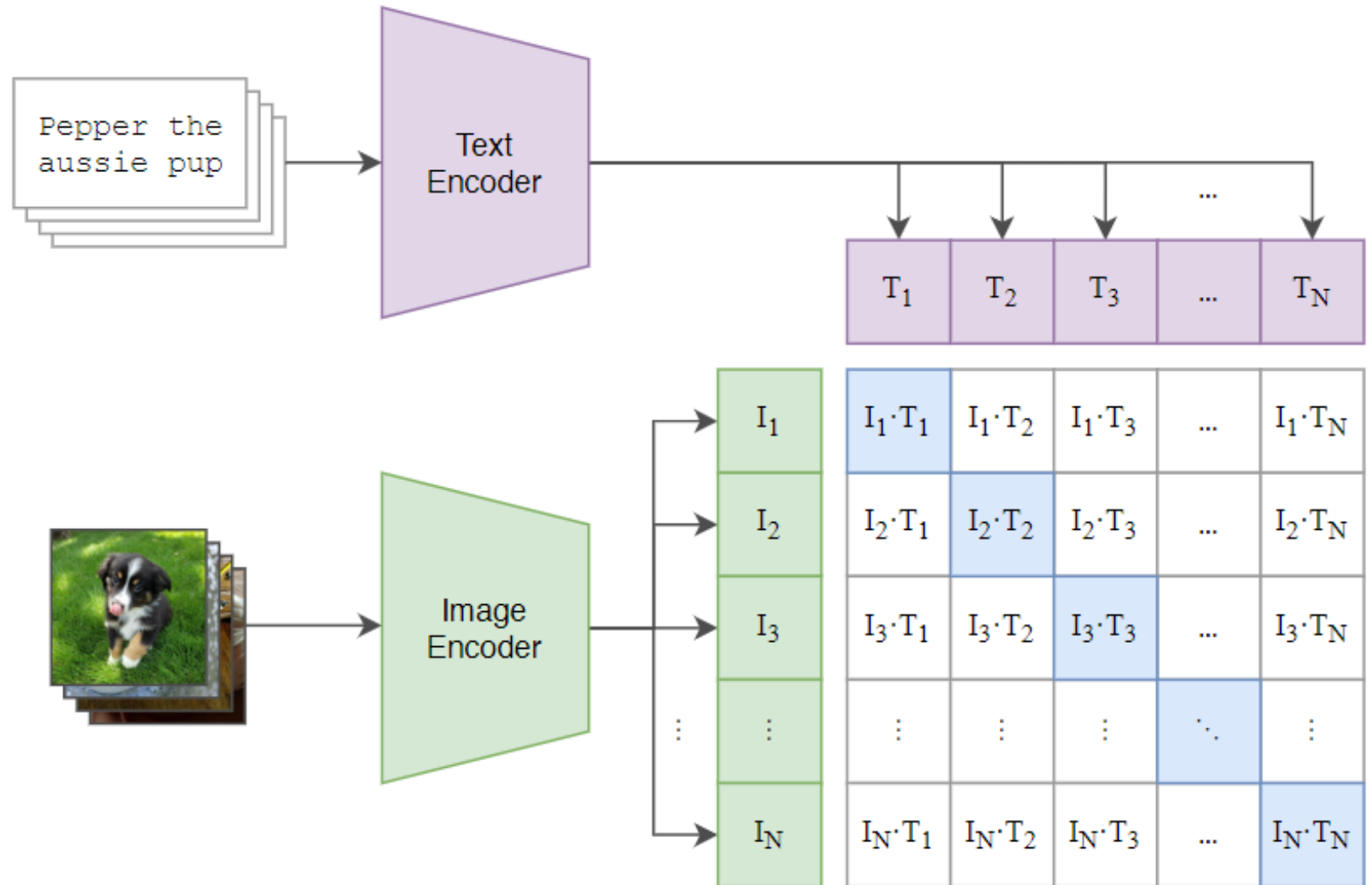
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Agenda

- Contexto
 - CLIP
 - Desafios
- *Visual Instruction Tuning*
- Arquitetura LLAVA

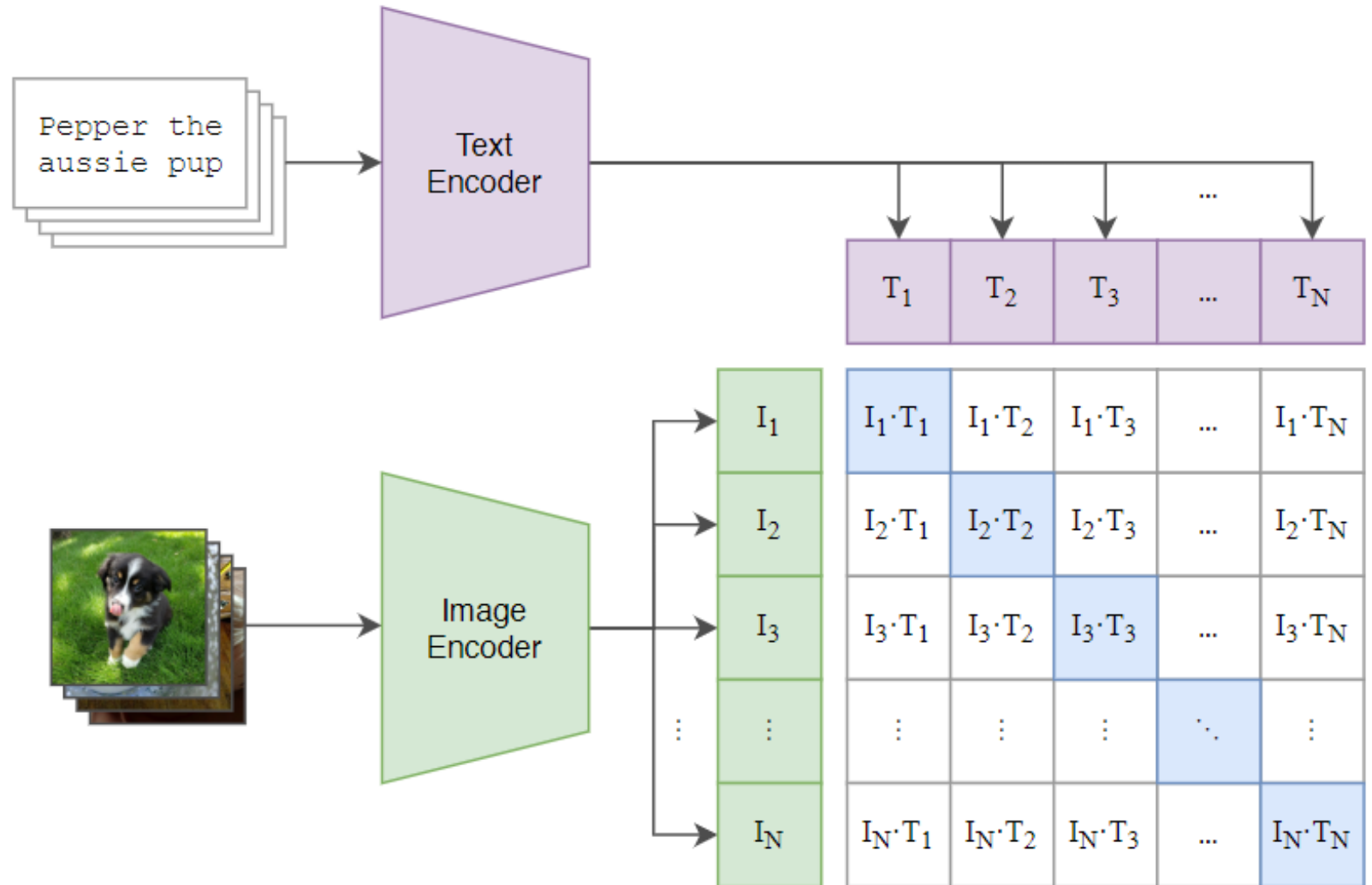
CLIP – *Contrastive Learning*

- O CLIP é formado um *encoder de texto* e um *encoder de imagem* que são treinados simultaneamente com objetivo de ter *embeddings similares*



CLIP – *Contrastive Learning*

- Computamos a similaridade do cosseno entre *embeddings* de texto e *embeddings* de imagem
- Em um *batch* de tamanho N vão existir N pares positivos para treinamento e $N^2 - N$ pares negativos



CLIP

```
# image_encoder - ResNet or Vision Transformer
# text_encoder  - CBOW or Text Transformer
# I[n, h, w, c] - minibatch of aligned images
# T[n, l]       - minibatch of aligned texts
# W_i[d_i, d_e] - learned proj of image to embed
# W_t[d_t, d_e] - learned proj of text to embed
# t             - learned temperature parameter

# extract feature representations of each modality
I_f = image_encoder(I) #[n, d_i]
T_f = text_encoder(T)  #[n, d_t]

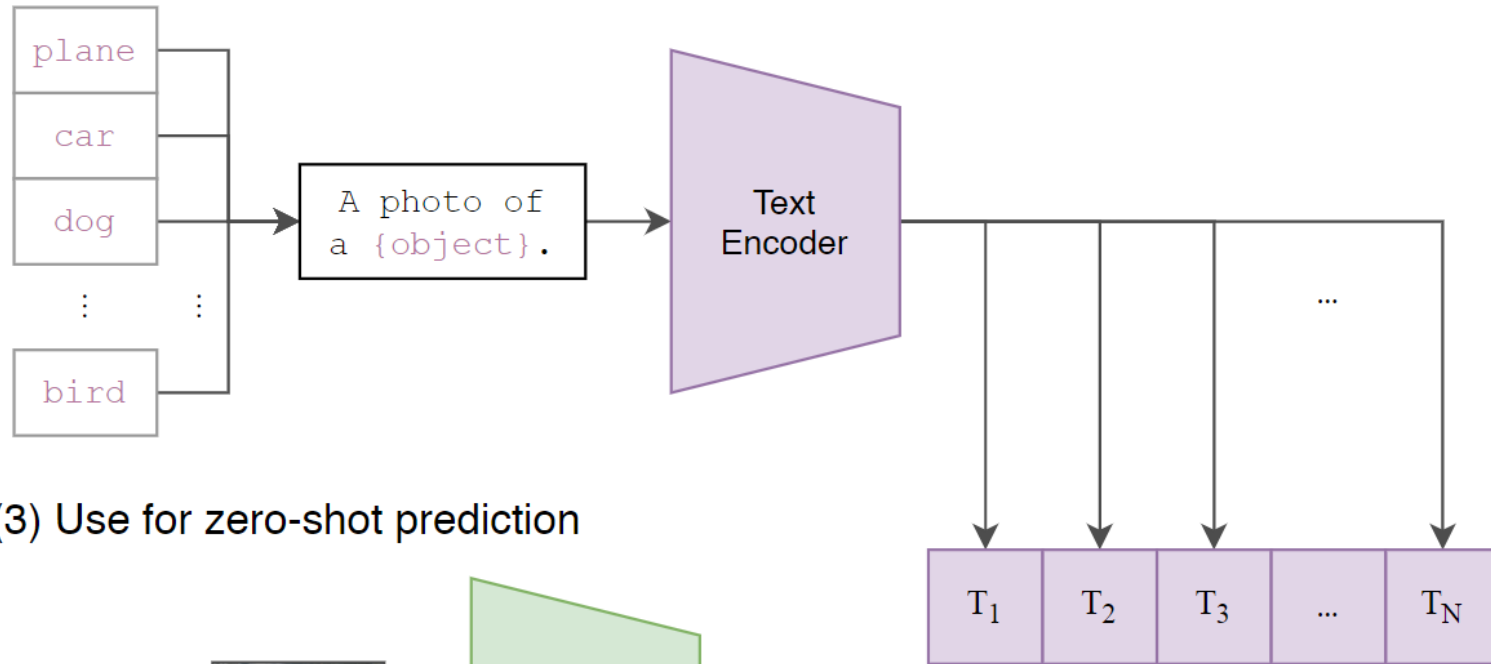
# joint multimodal embedding [n, d_e]
I_e = l2_normalize(np.dot(I_f, W_i), axis=1)
T_e = l2_normalize(np.dot(T_f, W_t), axis=1)

# scaled pairwise cosine similarities [n, n]
logits = np.dot(I_e, T_e.T) * np.exp(t)

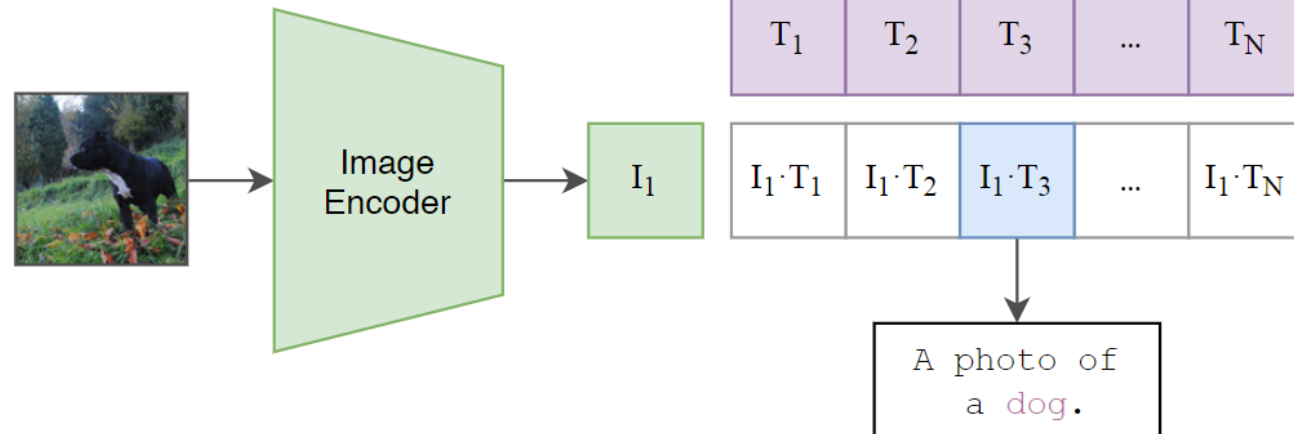
# symmetric loss function
labels = np.arange(n)
loss_i = cross_entropy_loss(logits, labels, axis=0)
loss_t = cross_entropy_loss(logits, labels, axis=1)
loss   = (loss_i + loss_t)/2
```

CLIP – *Contrastive Learning*

(2) Create dataset classifier from label text



(3) Use for zero-shot prediction



Desafios

- *Instruction Fine Tuning* era utilizado apenas para LLMs
- Apesar do CLIP ser um grande salto qualitativo para modelos multimodais, ainda não tínhamos nada equivalente a um assistente multimodal
- Principal problema é a ausência de um conjunto de dados *open source*

Visual Instruction Tuning

- Dataset CC3M - <https://huggingface.co/datasets/pixparse/cc3m-wds>
 - Imagem + Caption
- Como usar esses dados para criar um dataset de instruções?



man listening to music with
headphones on a street

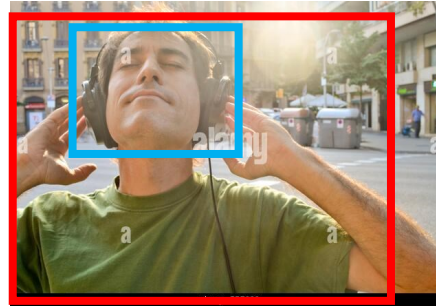
Visual Instruction Tuning

- Como usar esses dados para criar um dataset de instruções?

image



*Bounding box
annotation*



caption

man listening to music with
headphones on a street

context

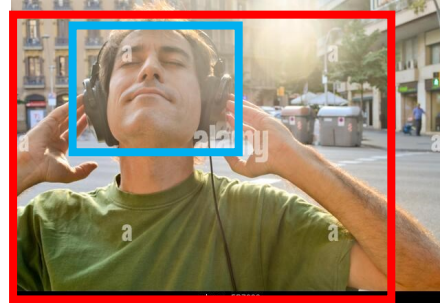
man [0.0 0.1 0.7 0.8]
headphone [0.2 0.15 0.4 0.25]

Visual Instruction Tuning

image



Bounding box
annotation



caption

man listening to music with
headphones on a street

context

man [0.0 0.1 0.7 0.8]
headphone [0.2 0.15 0.4 0.25]



ChatGPT

<SystemPrompt> You are an AI visual assistant that can analyze a single image. You receive five sentences, each describing the same image you are observing. In addition, specific object locations within the image are given, along with detailed coordinates. These coordinates are in the form of bounding boxes, represented as (x1, y1, x2, y2) with floating numbers ranging from 0 to 1. These values correspond to the top left x, top left y, bottom right x, and bottom right y.

Using the provided caption and bounding box information, describe the scene in a detailed manner.

Instead of directly mentioning the bounding box coordinates, utilize this data to explain the scene using natural language. Include details like object counts, position of the objects, relative position between the objects.

When using the information from the caption and coordinates, directly explain the scene, and do not mention that the information source is the caption or the bounding box. Always answer as if you are directly looking at the image.

<Prompt><Context1Captions> man listening to music with headphones on a street<Context2BoundingBoc>man [0.0 0.1 0.7 0.8] headphone [0.2 0.15 0.4 0.25]

<response> A man with a headphone on his head listening to music

Visual Instruction Tuning

Context type 1: Captions

A group of people standing outside of a black vehicle with various luggage.

Luggage surrounds a vehicle in an underground parking area

People try to fit all of their luggage in an SUV.

The sport utility vehicle is parked in the public garage, being packed for a trip

Some people with luggage near a van that is transporting it.

Context type 2: Boxes

person: [0.681, 0.242, 0.774, 0.694], backpack: [0.384, 0.696, 0.485, 0.914], suitcase: ...<omitted>

Response type 1: conversation

Question: What type of vehicle is featured in the image?

Answer: The image features a black sport utility vehicle (SUV) ...<omitted>

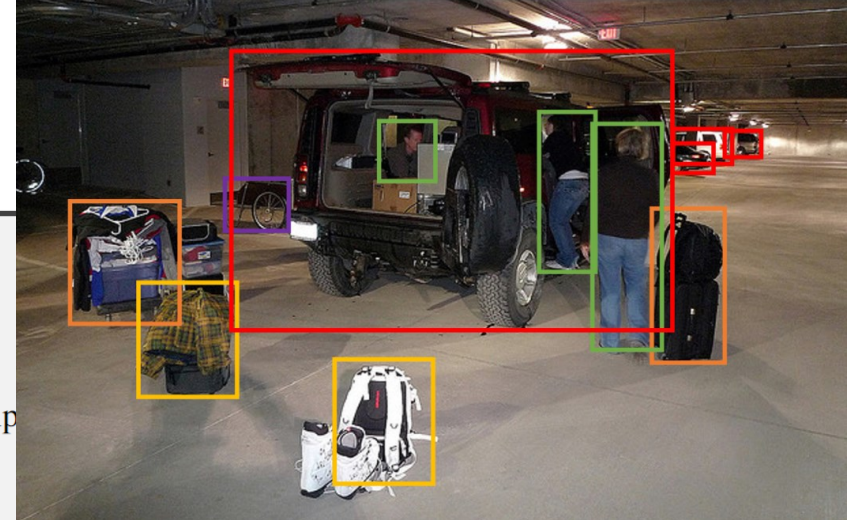
Response type 2: detailed description

The image is an underground parking area with a black sport utility vehicle (SUV) parked. There are three people in the scene, with one person standing closer to the left side of the vehicle, another person in the middle, and the third person on the right side. They are all working together to pack their luggage into the SUV for a trip. ...<omitted>

Response type 3: complex reasoning

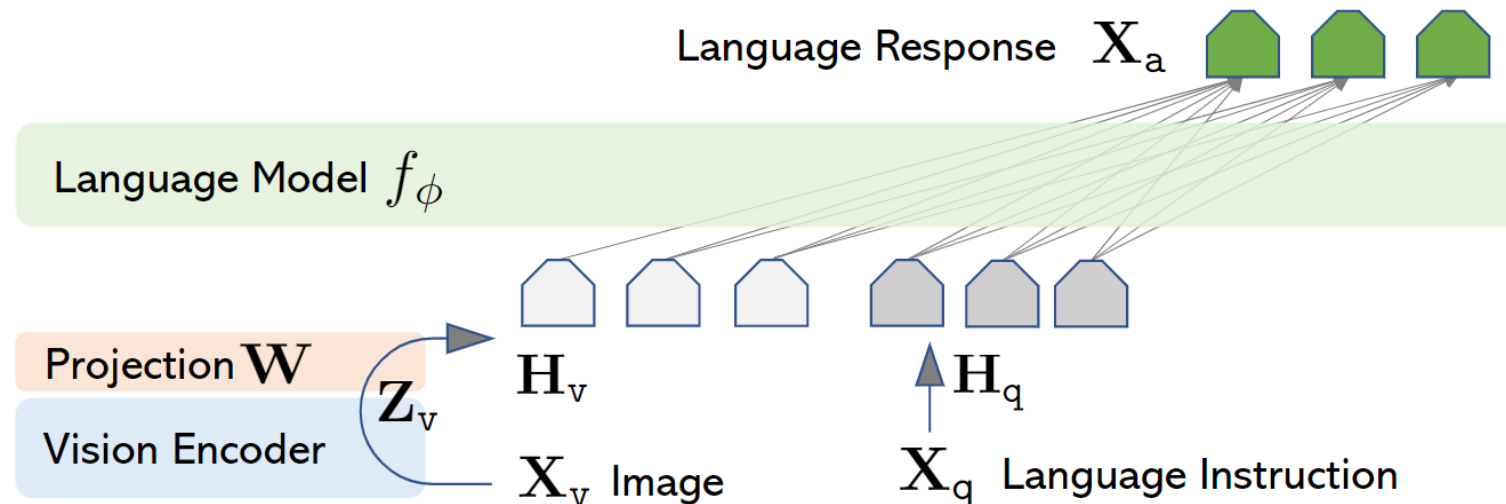
Question: What challenges do these people face?

Answer: In the image, a group of people is standing outside a black SUV in a parking area, surrounded by various pieces of luggage, including suitcases and backpacks. They are facing the challenge of fitting all their luggage into the black SUV. There are multiple suitcases and backpacks to be packed, which suggests that the group has a significant amount of belongings ...<omitted>



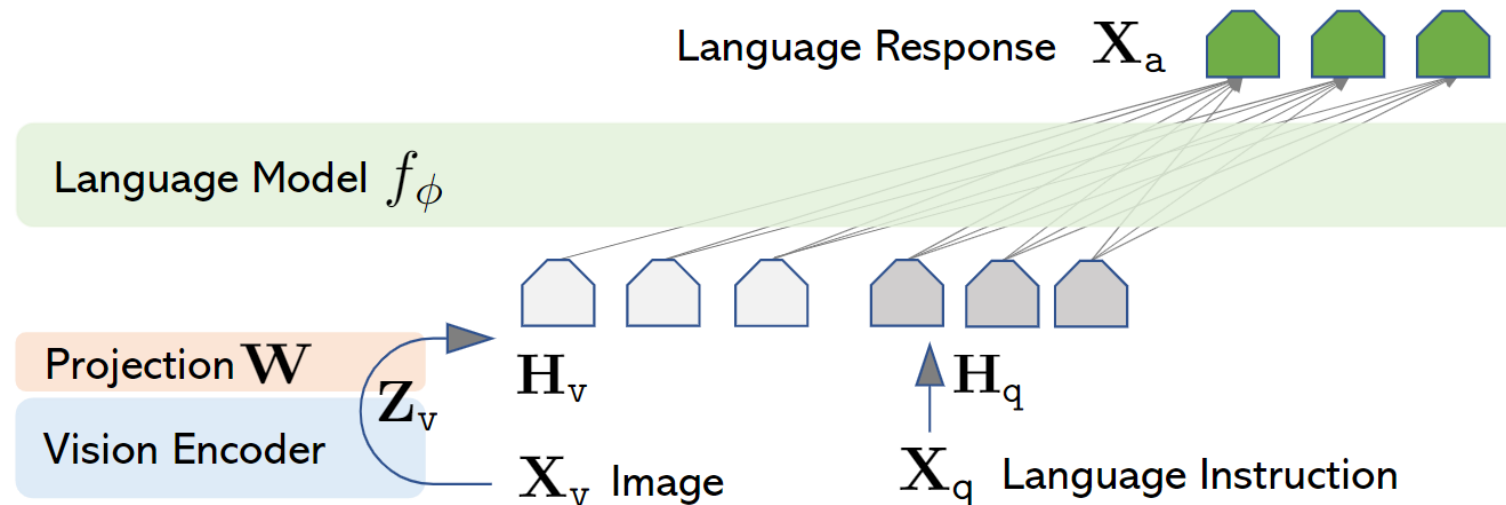
Arquitetura LLAVA

- Utiliza um LLM pré-treinado (Vicuna) para gerar a resposta
- Utiliza um *encoder* de imagens (OpenClip) também pré-treinado
- Uma camada linear (a partir do LLAVA 1.5 um MLP) para projetar a saída do *encoder* visual no espaço de embeddings do texto



Arquitetura LLAVA

- Treino em duas etapas
 - 1 Estágio– Apenas a matriz de projeção W é treinada com o prompt para descrever a imagem em detalhes, considerando o *ground truth* como o *caption* original
 - 2 Estágio – Tanto a matriz de projeção W quanto os pesos do LLM são ajustados no conjunto de dados gerado com o uso do ChatGPT.



Arquitetura LLaVA

Name		LLaVA-NeXT-7B	LLaVA-NeXT-13B	LLaVA-NeXT-34B
Model Size	Total	7.06B	13.35B	34.75B
	Vision Encoder	303.5M	303.5M	303.5M
	Connector	21M	31.5M	58.7M
	LLM	6.74B	13B	34.39B
Resolution		336 x [(2,2), (1,2), (2,1), (1,3), (3,1), (1,4), (4,1)]		
Stage-1	Training Data	558K		
	Trainable Module	Connector		
Stage-2	Training Data	760K		
	Trainable Module	Full model		
Compute (#GPU x #Hours)		8x20	16x24	32x30

Avaliando resultados

- É difícil de avaliar o quão boa é a resposta de um modelo
- Abordagens atuais:
 - Utilizar um modelo como juiz das respostas
 - Realizar ranqueamento a partir de respostas

```
[System]
Please act as an impartial judge and evaluate the quality of the responses provided by two
AI assistants to the user question displayed below. You should choose the assistant that
follows the user's instructions and answers the user's question better. Your evaluation
should consider factors such as the helpfulness, relevance, accuracy, depth, creativity,
and level of detail of their responses. Begin your evaluation by comparing the two
responses and provide a short explanation. Avoid any position biases and ensure that the
order in which the responses were presented does not influence your decision. Do not allow
the length of the responses to influence your evaluation. Do not favor certain names of
the assistants. Be as objective as possible. After providing your explanation, output your
final verdict by strictly following this format: "[[A]]" if assistant A is better, "[[B]]"
if assistant B is better, and "[[C]]" for a tie.

[User Question]
{question}

[The Start of Assistant A's Answer]
{answer_a}
[The End of Assistant A's Answer]

[The Start of Assistant B's Answer]
{answer_b}
[The End of Assistant B's Answer]
```

Figure 5: The default prompt for pairwise comparison.

```
[System]
Please act as an impartial judge and evaluate the quality of the response provided by an
AI assistant to the user question displayed below. Your evaluation should consider factors
such as the helpfulness, relevance, accuracy, depth, creativity, and level of detail of
the response. Begin your evaluation by providing a short explanation. Be as objective as
possible. After providing your explanation, please rate the response on a scale of 1 to 10
by strictly following this format: "[[rating]]", for example: "Rating: [[5]]".

[Question]
{question}

[The Start of Assistant's Answer]
{answer}
[The End of Assistant's Answer]
```

Figure 6: The default prompt for single answer grading.

LLAVA-Bench

Challenging examples from LLaVA-Bench (In-the-Wild):		
		
ICHIRAN Ramen [source]		Filled fridge [source]
Annotation	A close-up photo of a meal at ICHI-RAN. The chashu ramen bowl with a spoon is placed in the center. The ramen is seasoned with chili sauce, chopped scallions, and served with two pieces of chashu. Chopsticks are placed to the right of the bowl, still in their paper wrap, not yet opened. The ramen is also served with nori on the left. On top, from left to right, the following sides are served: a bowl of orange spice (possibly garlic sauce), a plate of smoke-flavored stewed pork with chopped scallions, and a cup of matcha green tea.	
Question 1	What's the name of the restaurant?	An open refrigerator filled with a variety of food items. In the left part of the compartment, towards the front, there is a plastic box of strawberries with a small bag of baby carrots on top. Towards the back, there is a stack of sauce containers. In the middle part of the compartment, towards the front, there is a green plastic box, and there is an unidentified plastic bag placed on it. Towards the back, there is a carton of milk. In the right part of the compartment, towards the front, there is a box of blueberries with three yogurts stacked on top. The large bottle of yogurt is Fage non-fat yogurt , and one of the smaller cups is Fage blueberry yogurt . The brand and flavor of the other smaller cup are unknown. Towards the back, there is a container with an unknown content.
Question 2	Describe this photo in detail.	What is the brand of the blueberry-flavored yogurt?
		Is there strawberry-flavored yogurt in the fridge?

Table 6: Challenging examples from LLaVA-Bench (In-the-Wild), we provide extremely-detailed annotation for each image for an accurate evaluation. Some questions require the model to extract details from high resolution image and to have a broad knowledge coverage.

Referências

- Paper original LLAVA - LIU, Haotian et al. Visual instruction tuning. Advances in neural information processing systems, v. 36, 2024.
- LLAVA 1.5 - LIU, Haotian et al. Improved baselines with visual instruction tuning. arXiv preprint arXiv:2310.03744, 2023.